ERC-RAG: An Energy-based Retrieval Clustering for RAG in Vietnamese Economy Chatbots

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Abstract. ReRetrieval-Augmented Generation (RAG) has emerged as an effective framework for improving the factual accuracy and reliability of Large Language Model (LLM)-based chatbots, particularly in domain-specific question-answering tasks. However, most existing RAG systems rely on cosine similarity over dense embeddings for document retrieval, which serves as a linear approximation of semantic relevance and may not adequately capture complex semantic relationships in specialized domains such as economic question answering.In this study, we propose an Energy-based Retrieval Clustering Retrieval-Augmented Generation (ERC-RAG) framework for Vietnamese economic chatbots. The proposed approach first encodes a user query into a dense vector representation and retrieves a set of candidate documents using cosine similarity. Subsequently, K-Means clustering is applied to organize the retrieved documents into semantic clusters, and energy distance is employed to measure the distributional similarity between the query embedding and each document cluster. The cluster with the smallest energy distance is selected to provide contextual information for answer generation.We evaluate the proposed framework on the Vietnamese Economy Information Database (VEID) using the Vietnamese Economic Question Answering Dataset (VNEIQAD) benchmark. Experimental results demonstrate that ERC-RAG consistently outperforms conventional RAG variants in retrieval effectiveness, producing responses that are more focused and numerically accurate. These findings provide empirical evidence that energy distance between query representations and document distributions is a promising approach for cluster-level retrieval in domain-specific RAG systems, particularly for low-resource languages such as Vietnamese.

Keywords: automatic question answering, chatbot, Vietnamese language, generative AI, retrieval-augmented generation, energy-based models, energy distance, KMeans clustering, retrieval clustering.

1 Introduction

Chatbots are software systems designed to simulate human conversations through text- or voice-based interactions. They have been applied across a wide range of specialized domains, including knowledge graph-based natural language understanding (Ait-Mlouk & Jiang, 2020) and intelligent healthcare consultation (Ni et al., 2024). In recent years, most high-performance chatbot systems have been built upon Large Language Models (LLMs), enabling coherent, context-aware, and fluent response generation.

Retrieval-Augmented Generation (RAG) has emerged as a widely adopted framework for improving the reliability and factual consistency of LLM-based systems.

Rather than relying solely on parametric knowledge encoded in model weights, RAG retrieves relevant documents from external knowledge sources and conditions the generation process on the retrieved context. This paradigm has been shown to mitigate hallucinations, improve answer correctness, and facilitate knowledge updates without requiring model retraining. Recent studies have also demonstrated the effectiveness of RAG for Vietnamese chatbot applications, particularly in domain-specific question-answering tasks (Ngo-Ho et al., 2024a; Ngo-Ho et al., 2024b).

Within document-based RAG systems, numerous retrieval enhancement techniques have been proposed to improve retrieval quality. These include maximal marginal relevance (MMR), which balances relevance and diversity among retrieved documents (Ye et al., 2023), and hybrid retrieval strategies that combine dense vector search with sparse keyword-based methods such as BM25 (Chen et al., 2023; Sawarkar et al., 2024). Despite these advances, most contemporary RAG retrievers continue to rely on cosine similarity over dense embeddings as the primary relevance scoring function. While computationally efficient, cosine similarity represents a relatively simple geometric measure and may not fully capture complex query–document relationships or task-specific semantic structures, particularly in domains requiring precise numerical and factual grounding, such as economic question answering.

This limitation is especially evident in Vietnamese economic chatbots, where even minor semantic mismatches or loosely related passages may lead to inaccurate interpretations of economic indicators, statistical figures, or financial information. Such challenges motivate the exploration of retrieval mechanisms that extend beyond conventional pairwise similarity measures while remaining compatible with practical RAG pipelines.

To address this issue, we propose the Energy-based Retrieval Clustering Retrieval-Augmented Generation (ERC-RAG) framework for Vietnamese economic chatbots. The proposed approach adopts a two-stage retrieval architecture. First, cosine similarity over a single query embedding is employed to efficiently retrieve a candidate document set. Next, the retrieved documents are grouped into semantic clusters using K-Means clustering, and energy distance is used to measure the distributional similarity between the query representation and each document cluster. The cluster exhibiting the smallest energy distance is selected as the contextual source for answer generation. This design enables retrieval decisions to be made at the cluster level, potentially capturing richer semantic relationships than conventional document-level ranking based solely on cosine similarity.

We conduct extensive experiments on the Vietnamese Economy Information Database (VEID) using the Vietnamese Economic Question Answering Dataset (VNEIQAD) benchmark. Experimental results demonstrate that ERC-RAG consistently improves retrieval effectiveness compared with widely used RAG baselines. These findings suggest that energy distance between a query representation and document cluster distributions constitutes a promising direction for cluster-level retrieval in domain-specific RAG systems, particularly for low-resource languages such as Vietnamese.

2 Energy-based concepts in LLMs and RAG

A typical RAG framework consists of a retrieval component that searches for external knowledge from structured or unstructured sources, including databases, books,

scientific articles, and web documents. Depending on the retrieval requirements and knowledge representation, different storage and retrieval mechanisms can be employed, such as vector databases, graph databases, and relational databases. In this work, we adopt a vector database to store and retrieve economic knowledge due to its compatibility with dense embedding-based retrieval. The investigation of alternative retrieval infrastructures is left for future research.

Recent studies have also explored the integration of Energy-Based Models (EBMs) with LLMs to enhance retrieval and generation performance. Originally introduced by LeCun et al. (2006), EBMs assign lower energy values to compatible input–output pairs and higher energy values to incompatible ones, providing a flexible framework for modeling relationships without requiring explicit probability normalization. Bhattacharyya et al. (2021) proposed an energy-based reranking mechanism for neural machine translation. Shankar et al. (2025) incorporated energy-based modeling into medical RAG systems to learn smooth energy landscapes over query–document pairs. Similarly, Cai et al. (2025) introduced Entriever, an energy-based retriever designed for knowledge-grounded dialogue systems.

Inspired by the concept of energy as a scalar potential, where desirable configurations correspond to lower energy values and mismatched configurations correspond to higher ones, we employ energy distance (Rizzo & Székely, 2016) as a retrieval criterion within the RAG pipeline. Unlike cosine similarity, which measures angular alignment between embedding vectors, energy distance provides a statistical measure of discrepancy between data distributions. This property enables a more informative comparison between a query representation and groups of retrieved documents viewed collectively as semantic clusters (Kayal et al., 2021).

Energy-Based Distance and Clustering Methods

Energy distance has been extensively studied as a statistical measure for quantifying discrepancies between probability distributions. Unlike cosine similarity, which evaluates angular similarity between individual vectors, energy distance captures distributional differences by aggregating pairwise distances among samples (Székely & Rizzo, 2013; Rizzo & Székely, 2016). This property makes it particularly suitable for comparing sets of embeddings rather than isolated vector representations. Previous research has explored the integration of energy distance into clustering algorithms. Notably, the k-groups framework generalizes K-Means by replacing centroid-based optimization with an objective derived from energy distance, enabling clustering based on distributional characteristics rather than Euclidean means alone (Li & Rizzo, 2017). In natural language processing, energy-based approaches have been applied to ranking and compatibility modeling tasks. Bhattacharyya et al. (2021) proposed an energy-based reranking method for neural machine translation. Cai et al. (2025) introduced Entriever, an energy-based retriever designed to model query–document compatibility in knowledge-grounded dialogue systems. Similarly, Shankar et al. (2025) investigated the integration of Energy-Based Models (EBMs) into Retrieval-Augmented Generation frameworks to improve reliability and uncertainty estimation.

Despite these advances, existing studies have primarily employed energy-based techniques for clustering, document-level reranking, or confidence estimation. The use of energy distance as a criterion for cluster selection within retrieval pipelines remains relatively underexplored. In particular, limited attention has been given to retrieval strategies that evaluate document clusters collectively with respect to a query representation, rather than relying solely on document-level relevance scores. This gap

motivates the proposed ERC-RAG framework, which leverages energy distance for cluster-level retrieval in domain-specific RAG systems.

Clustering for Retrieval-Augmented Generation

Clustering has emerged as an effective strategy for improving retrieval quality and contextual efficiency in Retrieval-Augmented Generation (RAG) systems. By grouping semantically related documents, clustering can reduce redundancy, improve information diversity, and alleviate context window limitations when interacting with Large Language Models (LLMs). Recent studies have proposed cluster-based retrieval frameworks to enhance RAG performance. These approaches typically organize retrieved documents into semantic clusters in order to compress contextual information and improve downstream reasoning by presenting only the most representative information from each cluster. Such strategies can reduce duplicated content while increasing the coverage of relevant knowledge within a limited context budget.

However, existing cluster-based retrieval methods generally rely on document-level similarity scores, such as cosine similarity, or heuristic aggregation strategies for cluster selection. As a result, clusters are often evaluated based on the relevance of individual documents rather than their collective semantic characteristics. Consequently, these approaches do not explicitly model the relationship between a query representation and an entire document cluster, potentially limiting their ability to capture higher-order semantic structures within the retrieved document set.

Reranking and Retrieval Optimization in RAG

Reranking is a fundamental component of modern information retrieval systems, aiming to refine the ordering of candidate documents following an initial retrieval stage. Traditional reranking approaches commonly rely on cross-encoder architectures that jointly encode query–document pairs and estimate their relevance scores (Nogueira & Cho, 2019). More recent studies have explored hybrid retrieval strategies that combine dense and sparse representations (Chen & Wiseman, 2023; Sawarkar et al., 2024), as well as LLM-based reranking methods that leverage the reasoning capabilities of large language models to improve retrieval quality (Sun et al., 2023).

Within Retrieval-Augmented Generation (RAG) systems, reranking is typically employed to improve the quality of retrieved contexts before answer generation. By reordering candidate documents according to estimated relevance, reranking can enhance the factual grounding and overall quality of generated responses. However, most existing reranking approaches operate at the level of individual documents and do not explicitly account for structural relationships among retrieved documents. As a result, potentially useful semantic information contained in groups of related documents may not be fully exploited during the retrieval process.

Research Gap

In summary, prior studies have investigated (i) energy distance for clustering and compatibility modeling, (ii) clustering techniques for improving retrieval efficiency and contextual organization, and (iii) reranking methods for refining document relevance in Retrieval-Augmented Generation systems.

However, limited research has explored the integration of these directions within a unified retrieval framework. In particular, the use of energy distance as a cluster selection criterion in RAG remains underexplored. Existing approaches generally rely on document-level similarity measures or heuristic aggregation strategies, rather than evaluating document clusters collectively with respect to a query representation. This gap motivates the proposed ERC-RAG framework, which introduces an energy-based

mechanism for assessing the relevance of document clusters and enables a more expressive cluster-level retrieval process for RAG systems.

3 ERC-RAG Framework for Vietnamese Economy Chatbots

Following the frameworks proposed by Ngo-Ho et al. (2024a, 2024b), we develop the Energy-based Retrieval Clustering Retrieval-Augmented Generation (ERC-RAG) framework for the Generative Vietnamese Economy Chatbot (GVEC). The proposed architecture extends a conventional RAG pipeline by incorporating two additional components: K-Means clustering over retrieved candidate documents and energy distance-based cluster scoring.

A conventional RAG pipeline begins by encoding user queries and documents stored in a vector database into dense vector representations using an embedding model. Retrieval is then performed by identifying semantically relevant documents through similarity measures, most commonly cosine similarity. The highest-ranked documents are selected as top-(k) candidates and incorporated into the prompt provided to the LLM. By grounding generation on externally retrieved knowledge, this approach helps mitigate hallucinations and improves factual consistency. However, document-level ranking based solely on cosine similarity may not adequately capture subtle semantic compatibility in specialized domains such as economic question answering.

The proposed method is motivated by the observation that cosine similarity primarily measures geometric proximity in the embedding space and may provide only a limited approximation of the underlying semantic compatibility between a query and candidate documents. To address this limitation while preserving retrieval efficiency, ERC-RAG adopts a two-stage retrieval architecture that combines single-query cosine-based candidate retrieval and energy-based cluster selection.

In the first stage, a user query (q) is encoded into a single embedding vector $(f(q))$. Cosine similarity is then employed to retrieve the top-(N) candidate documents from the vector database. This lightweight retrieval mechanism efficiently constructs a pool of potentially relevant documents for subsequent cluster-level analysis.

In the second stage, the retrieved candidate documents are grouped into semantically coherent clusters using k-Means clustering, where the optimal number of clusters is determined automatically using the silhouette score criterion. For each cluster, the energy distance is computed between the query representation

$$X = f(q)$$

and the set of document embeddings

$$Y = \{f(d) \mid d \in \text{cluster}\}$$

following the formulation of Rizzo and Székely (2016):

$$\mathcal{E}(X, Y) = 2\mathbb{E}\|X - Y\| - \mathbb{E}\|X - X'\| - \mathbb{E}\|Y - Y'\|$$

Pairwise distances between the query embedding and document embeddings are computed using Euclidean distance, following Kayal et al. (2021).

Since |X| = 1, the term E||X - X'|| = 0 vanishes from the formula. In practice, the energy distance is clipped to be non-negative. the self-distance term vanishes,

$$\mathbb{E}\|X - X'\| = 0$$

yielding the following empirical estimator:

$$\mathcal{E}(X,Y) = \frac{2}{|Y|}\sum_{y\in Y}\|f(q)-y\| - \frac{1}{|Y|^2}\sum_{y,y'\in Y}\|y-y'\|$$

The cluster with the minimum energy distance is selected, and the documents contained within that cluster are subsequently processed by an LLM-based relevance filtering module before being used as contextual evidence for answer generation. By evaluating document clusters collectively rather than ranking documents independently, the proposed energy-based scoring mechanism captures higher-order semantic structures within the retrieved candidate set and enables cluster-level retrieval based on the distributional characteristics of document embeddings.

From a retrieval perspective, ERC-RAG can be viewed as a cluster-aware retrieval framework integrating two complementary components: K-Means document clustering and energy distance-based cluster selection. Existing cluster-oriented retrieval approaches typically rely on document-level similarity scores or heuristic aggregation strategies. In contrast, ERC-RAG employs energy distance between a query representation and each document cluster as the cluster selection criterion. This formulation treats a document cluster as a sample-based representation of an underlying semantic distribution and leverages the theoretical properties of energy distance for measuring distributional discrepancy (Székely & Rizzo, 2013; Rizzo & Székely, 2016). The proposed framework therefore establishes a principled connection between cluster-level retrieval and statistical distribution matching in modern RAG systems.

Algorithm: Energy-based Retrieval Clustering RAG (ERC-RAG)
Input:
Query q
Document collection D
Embedding model f(·)
Pre-retrieval size N = 40
Output: Generated answer r
Step 1: Query Input
The system receives a text query from the user:

$$q \in \mathcal{Q}$$

Step 2: Query Embedding
The query is embedded using the multilingual E5 model (Wang et al., 2022) to produce a dense vector representation:

$$\mathbf{x} = f(q), X = \mathbf{x}$$

The query distribution X is a degenerate distribution consisting of a single point.
Step 3: Cosine Similarity Retrieval
The top-N candidate documents are retrieved from the vector database via cosine similarity:

$$\mathcal{C} = \mathrm{TopN}_{d\in D}\cos(f(q), f(d)), \qquad |\mathcal{C}| = N.$$

Step 4: Document Embedding Matrix
The embedding matrix of the retrieved candidate documents is constructed as:

$$\mathbf{V} = \left[f(d_1), f(d_2), \ldots, f(d_N)\right] \in \mathbb{R}^{N\times 768}$$

Step 5: Automatic Cluster Number Selection

For each clustering configuration $k \in \{2, \ldots, \min(10, N-1)\}$, KMeans is applied and the silhouette score is computed. The optimal number of clusters is selected as:

$$k^* = \arg\max_k \text{Silhouette}(k)$$

Step 6: Final KMeans Clustering

KMeans is applied using the optimal cluster number k*:

$$\{\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_{k^*}\} = \text{KMeans}(\mathbf{V}, k^*)$$

Each cluster groups semantically similar retrieved documents.

Step 7: Energy-Based Cluster Scoring

For each cluster $(\mathcal{C}_j)$, the energy distance between the query distribution $X = \{f(q)\}$ and the cluster document distribution $Y_j = \{f(d) \mid d \in \mathcal{C}_j\}$ is computed as:

$$\mathcal{E}(X, Y_j) = \frac{2}{|X||Y_j|} \sum_{x \in X} \sum_{y \in Y_j} \|x-y\| - \frac{1}{|X|^2} \sum_{x,x'} \in X \|x-x'\| - \frac{1}{|Y_j|^2} \sum_{y,y' \in Y_j} \|y-y'\|$$

Since |X| = 1, the term E||X - X'|| = 0 vanishes from the formula. In practice, the energy distance is clipped to be non-negative.

$$\mathcal{E}(X, Y_j) = \frac{2}{|Y_j|} \sum_{y \in Y_j} \|f(q) - y\| - \frac{1}{|Y_j|^2} \sum_{y,y' \in Y_j} \|y - y'\|$$

In practice, the energy distance is clipped to be non-negative.

Step 8: Cluster Selection

The cluster with the minimum energy distance is selected:

$$j^* = \arg\min_j \mathcal{E}(X, Y_j)$$

The corresponding document set is:

$$\mathcal{D}_{j^*} = \{\, d \mid d \in \mathcal{C}_{j^*} \,\}$$

Step 9: LLM-based Document Relevance Filtering

Documents in the selected cluster are evaluated by a Large Language Model (LLM) to determine whether they contain information relevant to the query. The filtered document set is:

$$\mathcal{D}_{\text{filtered}} = \{d \in \mathcal{D}_{j^*} \mid \text{LLMGrade}(q, d) = \text{yes}\}$$

Step 10: Response Generation

The filtered documents are used as context in the Retrieval-Augmented Generation framework:

$$r = \text{LLM}\big(\text{Prompt}(q, \mathcal{D}_{\text{filtered}})\big)$$

The generated response (r) is returned to the user.

4 Experiments

We evaluate the proposed ERC-RAG framework using the Vietnamese Economy Information Database (VEID), which was introduced in previous studies (Ngo-Ho et

al., 2024a; Ngo-Ho et al., 2024b). To the best of our knowledge, VEID is the first dataset specifically designed for the evaluation of chatbot systems in the Vietnamese economic domain. Ngo-Ho et al. (2025) further introduced a benchmark dataset consisting of 1,000 automatically generated question–answer pairs derived from VEID. This benchmark dataset is adopted in the present study. The experimental protocol follows the setup used in previous studies (Ngo-Ho et al., 2024a, 2024b, 2025), enabling direct comparison across different retrieval and generation configurations. To evaluate retrieval performance, we assess ranking quality using standard information retrieval metrics, including MRR, Hit@5, and NDCG@5.

Documents are segmented using the RecursiveCharacterTextSplitter with a chunk size of 600 characters and a chunk overlap of 80 characters, resulting in 120,165 indexed document chunks stored in ChromaDB. All queries and document chunks are encoded using the intfloat/multilingual-e5-base embedding model, which produces 768-dimensional L2-normalized embeddings and follows the E5 retrieval protocol through the use of the prefixes *query:* and *passage:*.

For candidate retrieval, cosine similarity is employed to retrieve the top 40 candidate documents for each query. The retrieved documents are subsequently grouped using K-Means clustering, where the number of clusters is selected from the range $k \in \{2, \ldots, \min(10, |C| - 1)\}$. The optimal value of (k) is determined automatically using the silhouette score criterion with random_state=42 and n_init='auto'.

Document grading is performed through a single batch LLM call applied to all documents within the selected cluster. The model returns a JSON array containing the indices of documents identified as relevant to the input query. The same LLM instance is used for both document grading and answer generation throughout all experiments. To provide a comprehensive evaluation, we report both generation-oriented and retrieval-oriented metrics. Generation quality is measured using ROUGE-2 (Recall, Precision, and F1) and BLEU-2 (Papineni et al., 2002). Semantic similarity between generated and reference answers is assessed using cosine similarity in the embedding space. Retrieval effectiveness is evaluated using Mean Reciprocal Rank (MRR), Hit@5, and Normalized Discounted Cumulative Gain at rank 5 (NDCG@5).

The proposed ERC-RAG framework achieved MRR = 0.62, Hit@5 = 0.64, and NDCG@5 = 0.83 on the VEID benchmark, indicating effective retrieval and ranking of relevant documents. The obtained NDCG@5 score further suggests that relevant documents are consistently positioned near the top of the ranked retrieval list.

Table 1 presents a detailed comparison between the proposed ERC-RAG framework, the Energy-based Reranking RAG baseline, and conventional RAG baselines on the VEID dataset. The evaluation includes both generation and retrieval metrics, namely ROUGE-2 (Recall, Precision, and F1), BLEU-2, Cosine Similarity, MRR, Hit@5, and NDCG@5. Conventional RAG baselines paired with VinaLLaMA-7B-chat, Vistral-7B-chat, OpenChat-3.5-7B, GPT-3.5-turbo, and GPT-4o-mini reported BLEU-2 scores ranging from 0.21 to 0.46. While the Energy-based Reranking RAG baseline additionally reported ROUGE-2 metrics, ERC-RAG further incorporates retrieval-oriented metrics such as Cosine Similarity, MRR, Hit@5, and NDCG@5, enabling a more comprehensive assessment of both retrieval effectiveness and ranking quality.

Table 1. Performance comparison of ERC-RAG, Energy-based Reranking RAG baseline, and conventional RAG models on ROUGE-2 Recall, ROUGE-2 Precision, ROUGE-2 F1, BLEU-2, Cosine Similarity, MRR, Hit@5, and NDCG@5 on the VEID dataset.

Algorithm	BLEU-2	ROUGE-2 Recall	ROUGE-2 Precision	ROUGE-2 F1	Cosine Similarity	Mean Reciprocal Rank	Hit@5	NDCG@5	Source
RAG + VinaLLaMA-7B-chat	0.26	-	-	-	-	-	-	-	(Ngo-Ho et al., 2025)
RAG + Vistral-7B-chat	0.23	-	-	-	-	-	-	-	(Ngo-Ho et al., 2025)
RAG + Openchat-3.5-7B	0.21	-	-	-	-	-	-	-	(Ngo-Ho et al., 2025)
RAG + GPT-3.5-turbo	0.46	-	-	-	-	-	-	-	(Ngo-Ho et al., 2025)
RAG + GPT-4o-mini	0.33	-	-	-	-	-	-	-	(Ngo-Ho et al., 2025)
Energy-based Reranking RAG + Grok-4-1-fast-reasoning	0.34	0.32	0.59	0.22	-	-	-	-	(Ngo-Ho et al., 2026)
Energy-based Reranking RAG + Gemini-2.5-flash-lite	0.52	0.5	0.69	0.4	-	-	-	-	(Ngo-Ho et al., 2026)
Energy-based Reranking RAG + GPT-4o-mini	0.61	0.57	0.74	0.49	-	-	-	-	(Ngo-Ho et al., 2026)
Our algorithms: ERC-RAG + GPT-5.0-mini	0.65	0.68	0.70	0.68	0.95	0.62	0.64	0.83	Our implementation

The ERC-RAG (single-vector) framework paired with GPT-5.0-mini (Row 9) achieved a ROUGE-2 F1 score of 0.68, ROUGE-2 Recall of 0.68, ROUGE-2 Precision of 0.70, and a BLEU-2 score of 0.65. Compared with the strongest Energy-based Reranking RAG configuration, ERC-RAG obtained higher ROUGE-2 F1 and BLEU-2 scores, with improvements of 0.19 and 0.04, respectively. The higher F1 score suggests a more balanced trade-off between answer coverage and precision, which may be attributed to the cluster-level retrieval mechanism employed by ERC-RAG.

In addition to generation metrics, ERC-RAG reports retrieval-oriented metrics that were not available for the baseline systems. A Cosine Similarity score of 0.95 indicates a high degree of semantic alignment between the selected retrieval context and the query. Furthermore, ERC-RAG achieved MRR = 0.62, Hit@5 = 0.64, and NDCG@5 = 0.83, suggesting that relevant documents are frequently ranked among the top retrieval results while maintaining strong ranking quality.

Overall, ERC-RAG achieved the highest reported scores among the evaluated configurations. These results suggest that combining semantic document clustering with energy distance-based cluster selection can improve both retrieval effectiveness

and response generation quality in domain-specific RAG systems. The findings further indicate that cluster-level retrieval based on energy distance represents a promising direction for economic question-answering applications in Vietnamese, where factual accuracy and semantic relevance are particularly important.

5 Conclusion

This paper presented the Energy-based Retrieval Clustering Retrieval-Augmented Generation (ERC-RAG) framework for domain-specific Vietnamese economic chatbots. Motivated by the limitations of cosine similarity in capturing complex query–document relationships, we proposed a two-stage retrieval architecture that combines cosine-based candidate retrieval with energy distance-based cluster selection. Specifically, candidate documents are first retrieved using a single query embedding and cosine similarity, after which K-Means clustering is applied to organize the retrieved documents into semantic clusters. Energy distance is then employed to measure the compatibility between the query representation and each document cluster, allowing the most relevant cluster to be selected as contextual evidence for answer generation.

Experimental results on the Vietnamese Economy Information Database (VEID) and the VNEIQAD benchmark indicate that the proposed framework improves retrieval effectiveness compared with conventional RAG baselines. These findings suggest that cluster-level retrieval based on energy distance provides a promising alternative to conventional document-level similarity ranking, particularly in domain-specific question-answering tasks where semantic precision is critical.

More broadly, this work contributes to the growing body of research investigating energy-based approaches for retrieval-augmented generation. While previous studies have primarily utilized energy functions for confidence estimation, document reranking, or abstention mechanisms, ERC-RAG explores the use of energy distance as a cluster selection criterion within a practical RAG pipeline. Future research directions include investigating alternative distributional similarity measures such as Maximum Mean Discrepancy (MMD) and Wasserstein distance, incorporating human feedback into the retrieval process, and improving computational efficiency to achieve a better balance among retrieval accuracy, response quality, and deployment cost in real-world applications.

References

1. Ait-Mlouk, A., & Jiang, L. (2020). KBot: a Knowledge graph based chatBot for natural language understanding over linked data. IEEE Access, 8, 149220-149230.
2. Chien Van Nguyen, Thuat Nguyen, Quan Nguyen, Huy Nguyen, Björn Plüster, Nam Pham, Huu Nguyen, Patrick Schramowski, Thien Nguyen. “Vistral-7B-Chat - Towards a State-of-the-Art Large Language Model for Vietnamese”. 2023.
3. Gutmann, M., & Hyvärinen, A. (2010). Noise-contrastive estimation: A new estimation principle for unnormalized statistical models. Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics (AISTATS).
4. Kayal, S. (2021, April). Unsupervised sentence-embeddings by manifold approximation and projection. In Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume (pp. 1-11).

5. Ngo-Ho, A.-K., Vo, K.-D., & Ngo-Ho, A.-K. (2024c). VQABG: Vietnamese question/answers benchmark generator for field-specific chatbot ground-truth dataset using EMINI (Exact Match wIth Numeric Information) indicator. CTU Journal of Innovation and Sustainable Development , 16(Special issue: ISDS), 80-90. https://doi.org/10.22144/ctujoisd.2024.325
6. Ngo-Ho, AK., Vo, KD., Ngo-Ho, AK. (2025). Evaluation of Large Language Models for the Vietnamese Language in Generative Vietnamese Economy Chatbots (GVEC) Services. In: Yeom, S. (eds) 6th International Conference on Electronics and Signal Processing. ICESP 2024. Signals and Communication Technology. Springer, Cham. https://doi.org/10.1007/978-3-031-94973-9_19
7. Ngo-Ho A.-Khoi., Ngo-Ho A.-Khoa, Vo K.-D. (2024a). GVEC: A Vietnamese Large Language Models Chatbot For Economy, Using Vietnamese Economy Information Database (VEID) From Vneconomy Community. The 7th International Conference on Multimedia Analysis and Pattern Recognition (MAPR 2024), Hanoi, Vietnam.
8. Ngo-Ho A.-Khoi., Ngo-Ho A.-Khoa, Vo K.-D. (2024b). GVEC: Generative Vietnamese Economy Chatbots using Vietnamese Numeric Economy Information Question/Answer Dataset (VNEIQAD) benchmark. The 17th International Conference on Advanced Technologies for Communications (ATC 2024), Hanoi, Vietnam.
9. Nguyen, Quan, Huy Pham, and Dung Dao. "VinaLLaMA: LLaMA-based Vietnamese Foundation Model". arXiv preprint arXiv:2312.11011, 2023.
10. Ni, P., Okhrati, R., Guan, S., & Chang, V. (2024). Knowledge graph and deep learning-based text-to-GraphQL model for intelligent medical consultation chatbot. Information Systems Frontiers, 26(1), 137-156.
11. Papineni, K., et al. (2002). Bleu: a method for automatic evaluation of machine translation. Proceedings of the 40th AMACL.
12. Rackauckas, Z. (2024). Rag-fusion: a new take on retrieval-augmented generation. arXiv preprint arXiv:2402.03367.
13. Sawarkar, K., Mangal, A., & Solanki, S. R. (2024). Blended rag: Improving rag (retriever-augmented generation) accuracy with semantic search and hybrid query-based retrievers. In 2024 IEEE 7th international conference on multimedia information processing and retrieval (MIPR) (pp. 155-161). IEEE.
14. Wang, Guan, et al. "Openchat: Advancing open-source language models with mixed-quality data". arXiv preprint arXiv:2309.11235, 2023.
15. Ye, X., Iyer, S., Celikyilmaz, A., Stoyanov, V., Durrett, G., & Pasunuru, R. (2023, July). Complementary explanations for effective in-context learning. In Findings of the Association for Computational Linguistics: ACL 2023 (pp. 4469-4484).
16. Bhattacharyya, S., Rooshenas, A., Naskar, S., Sun, S., Iyyer, M., & McCallum, A. (2021). Energy-based reranking: Improving neural machine translation using energy-based models. In Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics ACL 2021, (pp. 4528–4537). Association for Computational Linguistics. https://aclanthology.org/2021.acl-long.349
17. Cai, Y., Li, K., Huang, Y., Feng, J., & Ou, Z. (2025). Entriever: Energy-based retriever for knowledge-grounded dialog systems. In Findings of the Association for Computational Linguistics: ACL 2025 (pp. 21462–21474). Association for Computational Linguistics. https://aclanthology.org/2025.findings-acl.1105
18. Li, S., & Rizzo, M. L. (2017). *K-groups: A Generalization of K-means Clustering*. arXiv preprint arXiv:1711.04359. https://arxiv.org/abs/1711.04359
19. Chen, X., & Wiseman, S. (2023). BM25 query augmentation learned end-to-end. arXiv preprint arXiv:2305.14087. https://arxiv.org/abs/2305.14087
20. Khattab, O., & Zaharia, M. (2020). *ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT*. In Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval. https://arxiv.org/abs/2004.12832

21. LeCun, Y., Chopra, S., Hadsell, R., Ranzato, M., & Huang, F. (2006). A tutorial on energy-based learning. In G. Bakir, T. Hofmann, B. Schölkopf, A. Smola, B. Taskar, & S. V. N. Vishwanathan (Eds.), Predicting structured data (pp. 1–59). MIT Press. https://web.stanford.edu/class/cs379c/archive/2012/suggested_reading_list/documents/LeCunetal06.pdf
22. Nogueira, R., & Cho, K. (2019). Passage re-ranking with BERT. arXiv preprint arXiv:1901.04085. https://arxiv.org/abs/1901.04085
23. Rizzo, M. L., & Székely, G. J. (2016). Energy distance. Wiley Interdisciplinary Reviews: Computational Statistics, 8(1), 27–38. https://doi.org/10.1002/wics.1375
24. Shankar, R., Wong, S., Li, L., Bachmann, M., Silverthorne, A., Albert, B., & Jones, G. (2025). Energy landscapes enable reliable abstention in retrieval-augmented large language models for healthcare. arXiv preprint arXiv:2509.04482. https://arxiv.org/abs/2509.04482
25. Székely, G. J., & Rizzo, M. L. (2013). Energy statistics: A class of statistics based on distances. Journal of Statistical Planning and Inference, 143(8), 1249–1272. https://doi.org/10.1016/j.jspi.2013.03.018
26. Wang, L., Yang, N., Huang, X., Jiao, B., Yang, L., Jiang, D., Majumder, R., & Wei, F. (2022). *Text embeddings by weakly-supervised contrastive pre-training*. arXiv preprint arXiv:2212.03533. https://arxiv.org/abs/2212.03533
27. Sun, W., Yan, L., Ma, X.,Wang, S., Ren, P., Chen, Z., Yin, D., & Ren, Z. (2023). Is ChatGPT good at search? Investigating large language models as re-ranking agents. arXiv preprint arXiv:2304.09542. https://arxiv.org/abs/2304.09542
28. Asai, A., Wu, Z., Wang, Y., Sil, A., & Hajishirzi, H. (2023). Self-RAG: Learning to retrieve, generate, and critique through self-reflection. arXiv preprint arXiv:2310.11511. https://arxiv.org/abs/2310.11511